Towards AI-Driven Digital Twins and Intelligent Control for Accelerator Facilities

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Abstract

We present a preliminary framework linking physics-based digital twins, Bayesian optimization, reinforcement learning, and operator assistance for accelerator facilities. Digital-twin development targets the Upgraded Injector Test Facility at Jefferson Lab, combining component models with archived and live diagnostic data. In CEBAF simulations, a shared Gymnasium interface connects Xopt with the Scientific Optimization Control Toolkit (SoCT), enabling optimized initial states, replay-buffer seeding, and periodic Bayesian guidance during policy learning. One residual-control trace shows a reward increase of 3.7×10^{-5} above the Xopt handoff, with an intermediate decline below that baseline. Operator assistance combines logbook retrieval, source-linked summaries, and proposed Osprey orchestration. We outline the validation needed to assess predictive accuracy, control performance, and readiness for supervised facility use.

Keywords: Accelerator control, Digital twin, Bayesian optimization, Reinforcement learning, Machine learning, Beam diagnostics

1. Introduction

Particle accelerators must maintain reproducible operating conditions despite drift, configuration changes, and incomplete beam measurements. Tuning often requires repeated scans and invasive diagnostics, while detailed simulations can be too costly for direct use in feedback loops. Machine learning

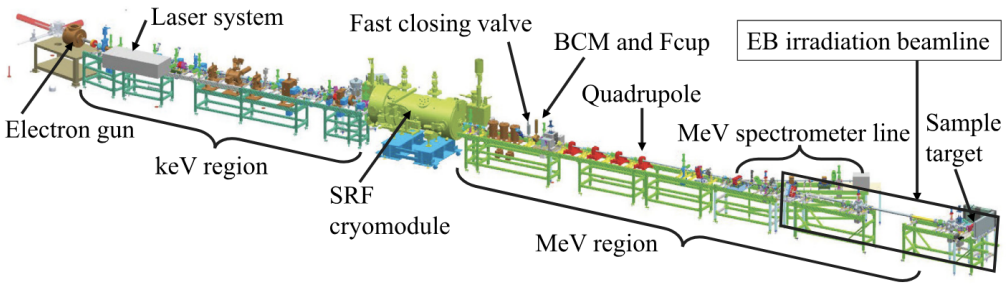


Figure 1: UITF layout showing the source, SRF cryomodule, transport elements, and diagnostics that define the proposed twin's scope. Layout from Li et al. [4].

can accelerate prediction and support control when models remain consistent with accelerator physics and their validated operating range [1].

A digital twin links physics models, diagnostic measurements, and the machine state to support prediction, calibration, and potentially non-invasive beam diagnostics. This shared representation can also support optimization and control. Bayesian optimization (BO) uses a surrogate model to select informative or promising settings when evaluations are expensive [2]; reinforcement learning (RL) learns a policy through repeated interaction with an environment. Combining them may allow policy learning to reuse information gathered during optimization.

This work develops interfaces among three components: an injector digital twin for the Upgraded Injector Test Facility (UITF), an Xopt–SoCT workflow tested in simulations of the Continuous Electron Beam Accelerator Facility (CEBAF), and operator assistance grounded in facility records. The contribution is their integration for accelerator tasks. A validated twin, improved control sample efficiency, and transfer between facilities remain development goals.

2. Digital-twin development for the UITF

2.1. Facility and modeling workflow

The UITF at Jefferson Lab is a compact superconducting radio-frequency accelerator used to test accelerator technologies and support low-energy experiments [3]. The proposed injector twin includes the electron source, RF cavities, magnetic transport elements, and diagnostics such as viewer screens (Fig. 1).

2.2. Validation

Validation will compare predicted and measured beam profiles and observables at known diagnostic locations. Training and test data should be separated by acquisition period and machine configuration to limit leakage between correlated records. Non-invasive inference will require comparison with independent diagnostics and a defined operating range. Development will proceed from baseline modeling and calibration to an integrated UITF prototype, followed by exploration of transfer to CEBAF.

3. Connecting Bayesian optimization and reinforcement learning

3.1. Shared environment and initialization

CEBAF cavity tuning involves multiple coupled controls subject to performance and operational constraints. To connect optimization with policy learning, we developed an adapter between Xopt [5] and the Gymnasium-based environment [6] used by the Scientific Optimization Control Toolkit (SoCT) [7]. Both tools evaluate settings in the same CEBAF simulation environment; the interface is intended to keep observations, bounds, and reward conventions consistent.

Xopt proposes absolute settings, whereas the CEBAF delta-action environment expects changes relative to the current state. The adapter converts each target to a delta action by subtracting the current setting. Evaluations record observations, rewards, termination status, and environment information for subsequent RL use.

The initial handoff used Xopt's best setting as the environment reset state. Compatible evaluation histories were also translated into state, action, reward, next-state, and termination tuples and inserted into the replay buffer of the Twin Delayed Deep Deterministic Policy Gradient (TD3) agent. This supplied prior transitions before the agent collected new experience.

3.2. Local refinement and periodic guidance

Starting from an optimized setting did not prevent TD3 exploration from moving away from it. Subsequent configurations restricted exploration magnitude and adjusted learning parameters to favor local refinement through delta actions. Training and inference rewards were examined separately to assess whether the learned policy retained or improved the initial performance.

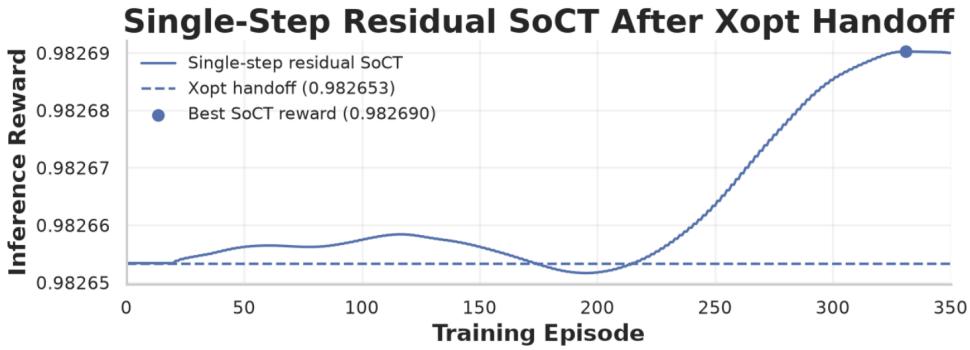


Figure 2: Residual SoCT inference reward versus training episode after an Xopt handoff. The narrow vertical range highlights the small variation around the baseline.

A later hybrid strategy introduced periodic Xopt searches during RL training. These searches supplied new targets to guide exploration while TD3 continued learning from environment interactions. Preliminary trials included 8- and 32-dimensional CEBAF environments; comparable repeated-run results are needed to assess how performance changes with dimensionality.

3.3. Preliminary reward observations

Figure 2 shows the residual-control trace, with an Xopt handoff reward of 0.982653 and a best displayed SoCT reward of 0.982690:

$$\Delta J = 0.982690 - 0.982653 = 3.7 \times 10^{-5}. \tag{1}$$

The trace initially remains near the handoff value, falls slightly below it, and later reaches its maximum. It therefore shows a small local improvement, without establishing monotonic improvement or reliable baseline preservation.

Figure 3 presents additional training and Xopt traces. Their objective definitions, uncertainty estimates, and relationship to the residual-control run are not specified. These plots alone do not establish statistical significance, improved sample efficiency, or a physical beam-performance gain. Evaluation should use repeated runs and a common interaction budget that includes BO evaluations, with the active handoff and guidance mechanisms specified for each comparison.

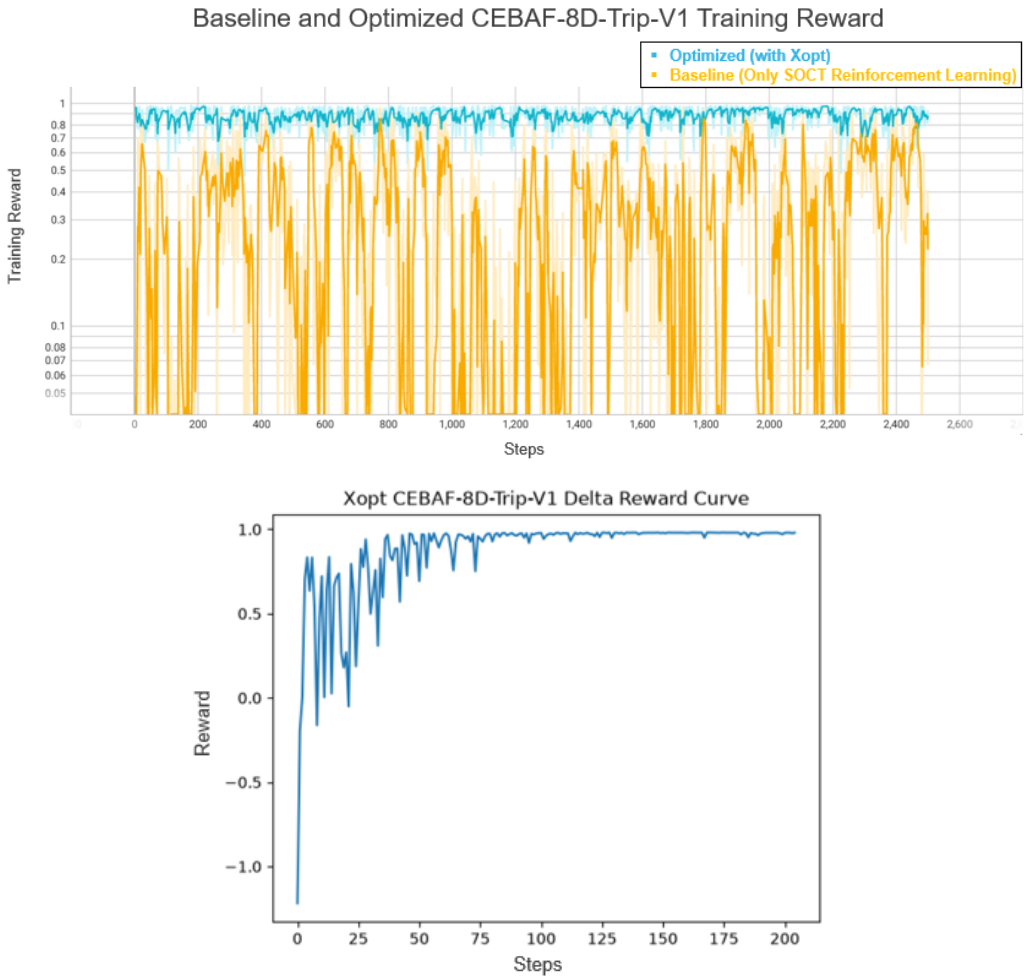


Figure 3: Top: training rewards for the curves labeled "Optimized (with Xopt)" and "Baseline (Only SoCT Reinforcement Learning)" in CEBAF-8D-Trip-V1, shown on a logarithmic reward axis. Bottom: the Xopt trace, titled "Delta Reward Curve" but labeled "Reward" on its vertical axis.

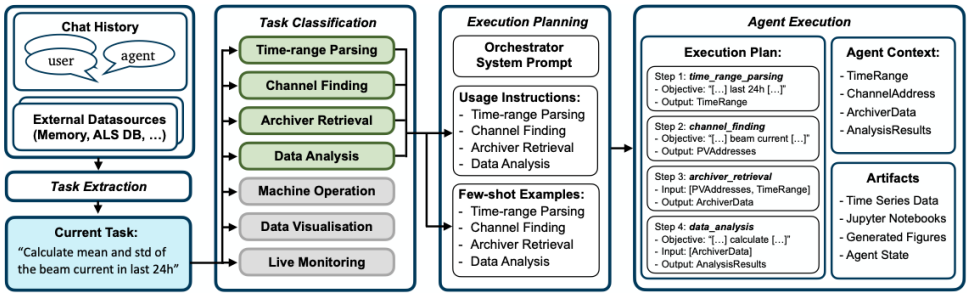


Figure 4: Osprey task extraction, classification, planning, and execution context. Reproduced from Fig. 2 of Hellert et al. [9] under CC BY 4.0. The diagram describes the external framework, rather than a fully deployed local integration.

4. Operator assistance

4.1. Logbook retrieval

The logbook workflow accepts a natural-language query and retrieves relevant records through read-only access to a database, API, or export. Records are normalized and filtered by available metadata, such as time, category, and severity. Ranking combines keyword matching with semantic similarity from sentence-transformer embeddings [8]; an LLM summarizes the retrieved material.

Responses are intended to include the top-k records with timestamps and identifiers so operators can inspect the supporting evidence. Embeddings are precomputed and cached, and searches operate on filtered subsets. Vector indexing is a possible extension; query rewriting is not yet included.

Evaluation is in progress and must assess both retrieval relevance and whether generated claims are supported by the cited records. Source links support traceability but do not guarantee factual accuracy or correct diagnosis, particularly when logs are incomplete.

4.2. Supervised orchestration

Osprey provides a framework for structured agentic interaction with control systems [9]. We are exploring its use to translate operator goals into classified tasks and execution plans (Fig. 4). Read-only log retrieval is the initial use case. Machine actions would require explicit authorization, parameter validation, and a record of the approved action and outcome.

5. Physics-aware checks

A proposed check layer would compare requested settings with permitted bounds and assess model predictions against measurements and known physical relationships. These checks must be tailored to each subsystem and tested under normal drift and deliberately inconsistent inputs. Detection performance has not yet been established. Human review and existing facility protection systems remain essential; model agreement alone cannot establish that an action is safe.

6. Conclusions and outlook

The framework connects UITF digital-twin development, BO-assisted policy learning in CEBAF simulations, and source-grounded operator assistance. The Xopt–SoCT interface supports optimized initialization, replay-buffer seeding, and periodic BO guidance. Early trials showed that initialization alone did not reliably retain the BO solution, motivating constrained local exploration and continued optimization guidance. The reported residual trace shows only a small local reward improvement.

Next steps are independent UITF model validation, controlled BO–RL comparisons across repeated runs and control dimensions, and evaluation of retrieval accuracy and physics-aware checks. These studies are needed before claims of improved efficiency, facility transfer, or operational readiness can be made.

Acknowledgments

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